Eingänge der Portexpander

C50 100nF +3V3
C51 100nF +3V3
C52 100nF +3V3
C53 100nF +3V3

Y7G VSS 8 VDD 1
Y9G VSS 8 VDD 1
Y11G VSS 8 VDD 1
Y10G VSS 8 VDD 1

GND

R32 100k	X4Z	Y7A 3 2 4050D	FPGA_IN_G1
R33 100k	X4AP	Y7B 5 4 4050D	FPGA_IN_K1
R34 100k	X4K	Y7C 7 6 4050D	FPGA_IN_L1
R35 100k	X4L	Y7D 9 10 4050D	FPGA_IN_I1
R36 100k	X4J	Y7E 11 12 4050D	FPGA_IN_L3
R37 100k	X4AA	Y7F 14 15 4050D	FPGA_IN_I3
R38 100k	X4H	Y9D 9 10 4050D	FPGA_IN_L2
R39 100k	X4AN	Y9E 11 12 4050D	FPGA_IN_I2
R40 100k	X4Y	Y9B 5 4 4050D	FPGA_IN_G3
R41 100k	X4AO	Y9C 7 6 4050D	FPGA_IN_K3
R42 100k	X4I	Y9A 3 2 4050D	FPGA_IN_G2
R43 100k	X4X	Y9F 14 15 4050D	FPGA_IN_K2
R50 22k	X4AJ	Y11F 14 15 4050D	FPGA_IN_UC_A3
R51 22k	X4D	Y11E 11 12 4050D	FPGA_IN_UC_A4
R52 22k	X4AM	Y11B 5 4 4050D	FPGA_IN_UC_V_KARTE
R53 22k	X4W	Y11A 3 2 4050D	FPGA_IN_UC_TUER
R54 22k	X4AK	Y11D 9 10 4050D	FPGA_IN_UC_V_VMIN
R55 22k	X4U	Y11C 7 6 4050D	FPGA_IN_UC_MG
R44 100k	X4AS	Y10A 3 2 4050D	FPGA_IN_G4
R45 100k	X4M	Y10B 5 4 4050D	FPGA_IN_K4
R46 100k	X4AC	Y10C 7 6 4050D	FPGA_IN_L4
R47 100k	X4AB	Y10D 9 10 4050D	FPGA_IN_I4
R48 22k	X4G	Y10E 11 12 4050D	FPGA_IN_UC_A1
R49 22k	X4E	Y10F 14 15 4050D	FPGA_IN_UC_A2

Ausgänge AV-Sperre und Z6

C61 100nF
Y12B GND 10 VCC 20
+5V

10k R56 10k R57 10k R58 10k R59

Y12A 54LS245N

FPGA_OUT_A1	2 A1	B1 18
FPGA_OUT_A2	3 A2	B2 17
FPGA_OUT_A3	4 A3	B3 16
FPGA_OUT_A4	5 A4	B4 15
FPGA_OUT_Z6	6 A5	B5 14
FPGA_OUT_P13	7 A6	B6 13
+5V	8 A7	B7 12
	9 A8	B8 11
	1 DIR	
	19 G	

X4T X4AI X4C X4S X4AU X4AG

AV-Sperre A1-A4

7-Segment Anzeigen

X10 5-104186-5
X11 5-104186-5

CON_7SEG_LI_IA	7 IA	A 13
CON_7SEG_LI_IB	1 IB	B 12
CON_7SEG_LI_IC	2 IC	C 11
CON_7SEG_LI_ID	6 ID	D 10
		E 9
	3 LT	F 15
	4 BI	G 14
	5 LE	

Y13A 4511N
Y14A 4511N

R60 1k R61 1k R62 1k R63 1k R64 1k R65 1k R66

CON_7SEG_RE_IA	7 IA	A 13
CON_7SEG_RE_IB	1 IB	B 12
CON_7SEG_RE_IC	2 IC	C 11
CON_7SEG_RE_ID	6 ID	D 10
CON_7SEG_5V	3 LT	E 9
	4 BI	F 15
CON_7SEG_GND	5 LE	G 14
CON_7SEG_LE		

R67 1k R68 1k R69 1k R70 1k R71 1k R72 1k R73

VDD 16 Y13B VSS 8
100nF C66
VDD 16 Y14B VSS 8
100nF C67

DS2 SC03-12EWA

14	ANODE_A	ANODE_A	14
13	ANODE_B	ANODE_B	13
12	COMMON_CATHODE_2	COMMON_CATHODE_2	12
11	NO_PIN_4	NO_PIN_4	11
10	NO_PIN_3	NO_PIN_3	10
9	ANODE_DP2	ANODE_DP2	9
8	ANODE_C	ANODE_C	8
1	ANODE_F	ANODE_F	1
2	ANODE_G	ANODE_G	2
3	NO_PIN_1	NO_PIN_1	3
4	COMMON_CATHODE_1	COMMON_CATHODE_1	4
5	NO_PIN_2	NO_PIN_2	5
6	ANODE_E	ANODE_E	6
7	ANODE_D	ANODE_D	7

DS1 SC03-12EWA

1	ANODE_F	ANODE_A	14
2	ANODE_G	ANODE_B	13
3	NO_PIN_1	COMMON_CATHODE_2	12
4	COMMON_CATHODE_1	NO_PIN_4	11
5	NO_PIN_2	NO_PIN_3	10
6	ANODE_E	ANODE_DP2	9
7	ANODE_D	ANODE_C	8

Sheet: